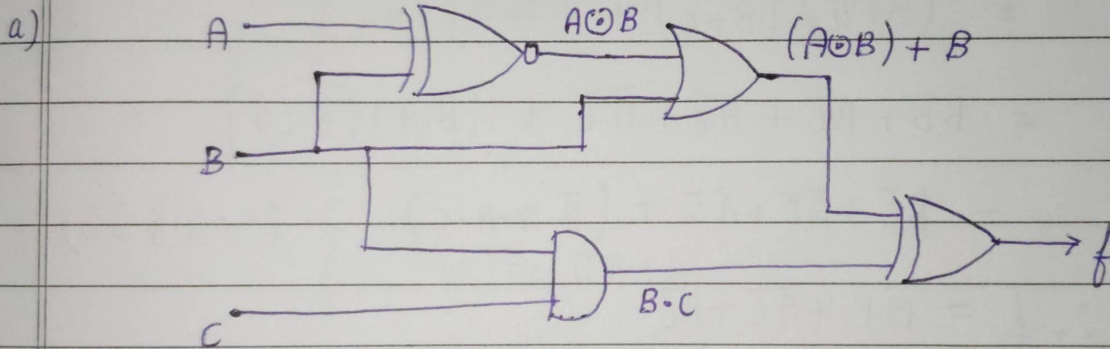


Day 2

Boolean Algebra - Assignment

- 1) Derive the output expression 'f' for the below circuits and draw the simplified $\square$ circuits.



$\Rightarrow$ From circuit,

$$f = ((A \odot B) + B) \oplus (B \cdot C)$$

Since, $A \odot B = AB + \bar{A}\bar{B}$

$$f = (AB + \bar{A}\bar{B} + B) \oplus (B \cdot C)$$

$$= (B(A+1) + \bar{A}\bar{B}) \oplus (B \cdot C)$$

$$= (B + \bar{A}\bar{B}) \oplus (B \cdot C)$$

$$= ((B + \bar{A})(B + \bar{B})) \oplus (B \cdot C)$$

$$f = (B + \bar{A}) \oplus (B \cdot C)$$

$\therefore A \oplus B = A\bar{B} + \bar{A}B$

$$\Rightarrow f = (B + \bar{A})(\overline{B \cdot C}) + (\overline{B + \bar{A}})(B \cdot C)$$

$$= (B + \bar{A})(\bar{B} + \bar{C}) + (\bar{B} \cdot \bar{A})(B \cdot C)$$

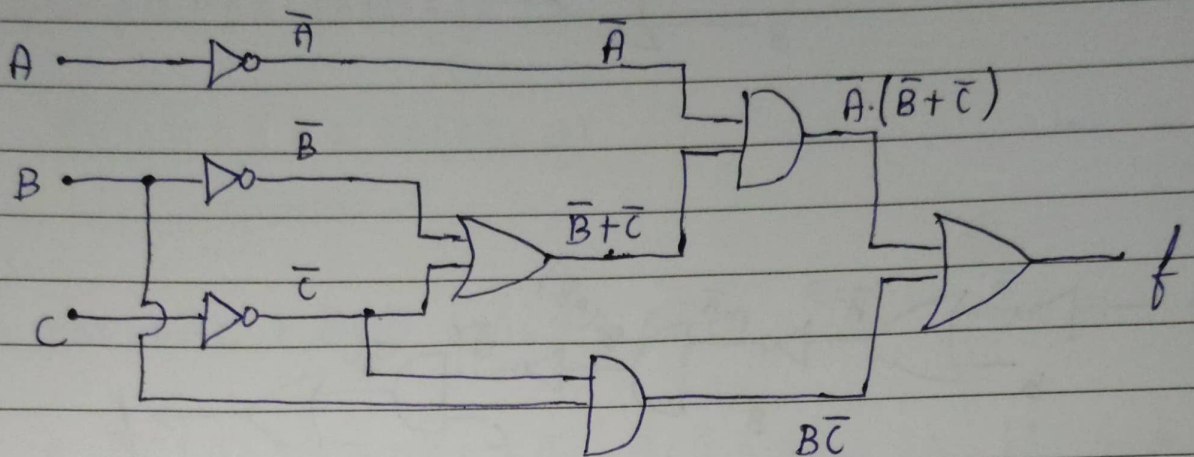
$$= B\bar{B} + B\bar{C} + \bar{A}\bar{B} + \bar{A}\bar{C} + ((\bar{B} \cdot \bar{A})(B \cdot C))$$

$$= B\bar{C} + \bar{A}\bar{B} + \bar{A}\bar{C} + (\bar{B} \cdot \bar{A} \cdot B \cdot C) \quad (\because B \cdot \bar{B} = 0)$$

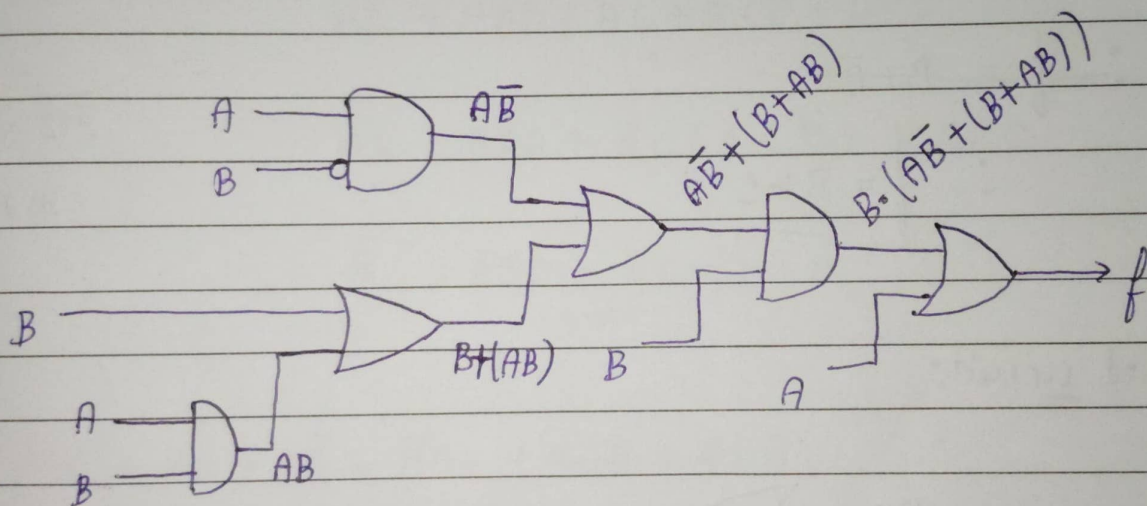
$$\therefore f = \bar{A}\bar{B} + \bar{A}\bar{C} + B\bar{C}$$

$$f = \bar{A}(\bar{B} + \bar{C}) + B\bar{C}$$

Simplified Circuit:



b)



$\Rightarrow$

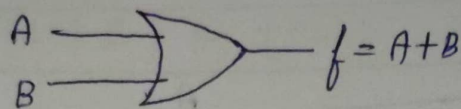
$$f = A + (B \cdot (\bar{A}B + (B + AB)))$$

$$= A + (B \cdot (\bar{A}B + B))$$

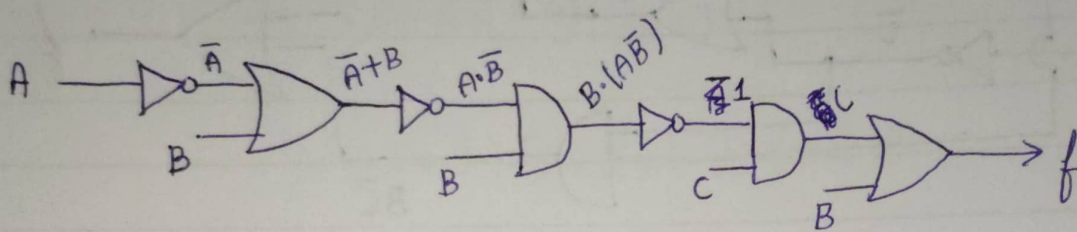
$$f = A + (B \cdot B)$$

$$\therefore f = \underline{\underline{A + B}}$$

Simplified circuit:



c)



$\Rightarrow$

$$\therefore f = \underline{\underline{B + AC}}$$

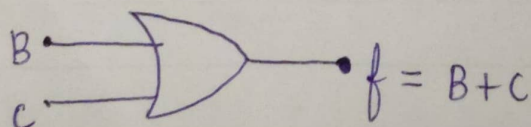
$$\therefore f = \underline{\underline{B + C}}$$

$$\overline{A+B} = A \cdot \bar{B}$$

$$B = A \cdot \bar{B}$$

A

Simplified circuit:



2) Minimize the following Boolean Expression using Boolean postulates.

a) $\bar{A}C\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

$$\Rightarrow \bar{A}C\bar{D}E + \bar{A}\bar{B}\bar{D} + AB(CE + D)$$

$$\therefore \bar{A}\bar{D}(CE + \bar{B}) + AB(CE + D)$$

b) $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

$$\Rightarrow \bar{B}\bar{C}(\bar{A} + A) + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D \quad (\because A + \bar{A} = 1)$$

$$\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D$$

$$\bar{B}\bar{C} + ABD + \bar{A}C(1 + \bar{D}) + A\bar{C}D \quad (\because 1 + \bar{D} = 1)$$

$$\bar{B}\bar{C} + ABD + \bar{A}C + A\bar{C}D$$

c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$

$$\Rightarrow \bar{A}\bar{C} + BC(1 + \bar{A}) + A\bar{B} + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$$

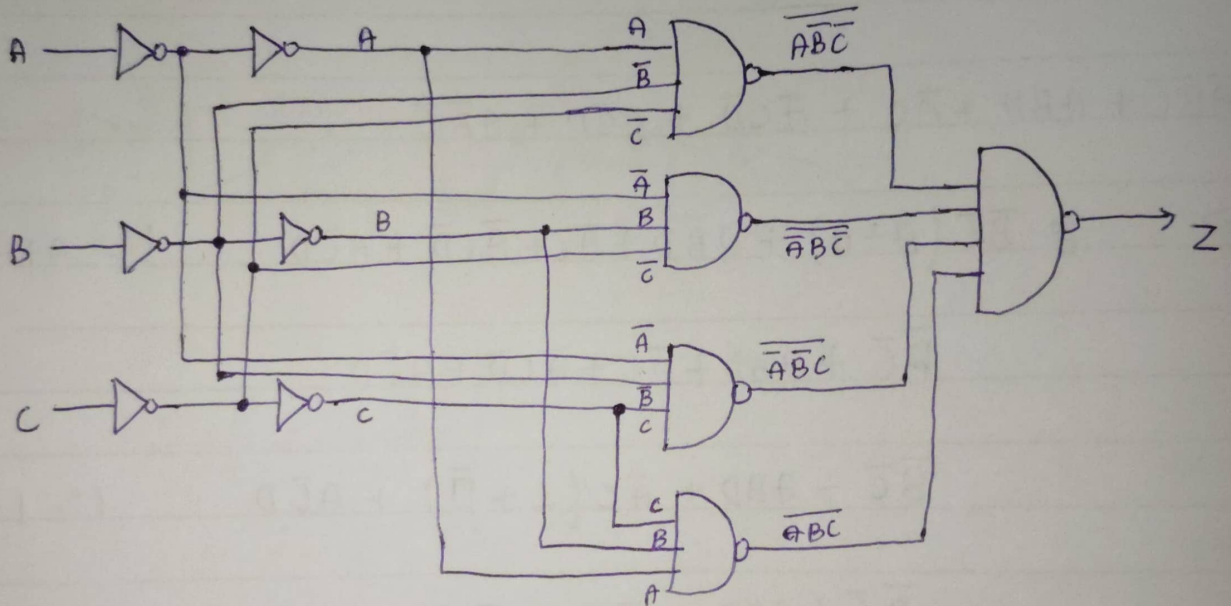
$$\bar{A}\bar{C} + BC + A\bar{B} + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$d) \bar{X}_1 \bar{X}_3 \bar{X}_4 + X_3 X_4 + \bar{X}_1 \bar{X}_2 X_4 + X_1 X_2 \bar{X}_3 X_4 \cancel{+ X_2} + X_2$$

$$\Rightarrow \bar{X}_1 \bar{X}_3 \bar{X}_4 + X_3 X_4 + \bar{X}_1 \bar{X}_2 X_4 + X_2 \quad (\because 1 + A = 1)$$

$$\therefore \bar{X}_1 \bar{X}_3 \bar{X}_4 + X_3 X_4 + \bar{X}_1 X_4 + X_2 \quad (\because X + \bar{X}Y = X + Y)$$

3) Which single gate can represent the functionality of given circuit?



$$\Rightarrow Z = (\overline{ABC}) (\overline{ABC}) (\overline{ABC}) (\overline{ABC})$$

$$= (\overline{ABC}) + (\overline{ABC}) + (\overline{ABC}) + (\overline{ABC})$$

$$= \bar{C}(A\bar{B} + \bar{A}B) + C(AB + \bar{A}\bar{B})$$

$$= \bar{C}(A \oplus B) + C(A \odot B)$$

$$\therefore \overline{A \oplus B} = A \odot B$$

$$\Rightarrow Z = \overline{C} (A \oplus B) + C (\overline{A \oplus B})$$

$$Z = (A \oplus B) \oplus C$$

$$\therefore Z = \underline{\underline{A \oplus B \oplus C}}$$

$\therefore$ the single gate representing the circuit is a 3-input XOR gate.

4) Minimize the following Boolean Expressions using K-map.

a) $f(a, b, c, d) = \sum_m (4, 6, 8, 10, 11, 12, 15) + \sum_d (3, 5, 7, 9)$

$\Rightarrow$

cd \ ab	$\overline{a}\overline{b}$	$\overline{a}b$	$a\overline{b}$	ab
$\overline{c}\overline{d}$	0 0	0 1	X 5	0 4
$\overline{c}d$	0 3	X 7	X 6	0 2
$c\overline{d}$	1 12	0 13	1 15	0 14
cd	1 8	X 9	1 11	1 10

$$f = \overline{a}b + a\overline{b} + cd + \overline{a}\overline{c}\overline{d}$$

$$\therefore f = (a \oplus b) + \underline{\underline{cd + \overline{a}\overline{c}\overline{d}}}$$

b) $f(a, b, c, d) = \prod_m(0, 2, 4, 6, 7, 9) \cdot \prod_d(10, 11)$

$\Rightarrow$

	$c+d$	$c+d$	$\bar{c}+\bar{d}$	$\bar{c}+\bar{d}$
$a+b$	0	1	1	0
$a+b$	0	1	0	0
$\bar{a}+\bar{b}$	1	1	1	1
$\bar{a}+b$	1	0	X	X

$\therefore f = (a+d) \cdot (a+\bar{b}+\bar{c}) \cdot (\bar{a}+b+\bar{d})$